Benthic Macroinvertebrate Community Change Following the 2025 Palisades Fire in Topanga Lagoon

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Abstract

The January 2025 Palisades Fire burned across the Topanga Creek watershed upstream of Topanga Lagoon, designated critical habitat for the federally endangered northern tidewater goby (*Eucyclogobius newberryi*). We compared benthic macroinvertebrate (BMI) assemblages before and after the fire to characterize early community change and its implications for goby prey availability. Substrate samples collected on six dates between May 2024 and February 2026 were preserved in 70% ethanol, then sorted, identified, and counted under magnification. Taxon richness fell from 20 taxa before the fire to as few as 2 in early post-fire samples, and Shannon diversity (H′) declined from 1.74 to 0.34. Ostracods dominated the early post-fire assemblage, while Hydropsychidae, present before the fire, were not recorded in any post-fire sample through February 2026. BMI and goby survey trends followed similar trajectories: sharp post-fire declines, then increases beginning in fall 2025. These results document a severe simplification of the lagoon's invertebrate community and incomplete recovery more than a year after the fire. Because the dataset includes one pre-fire sample and no reference site, the findings are descriptive and do not isolate fire as the sole cause. We also outline a GIS-based follow-up study that will evaluate field-verified goby occurrences beyond the mapped critical habitat unit using channel slope, passage barriers, and flow conditions.

Keywords: Topanga Lagoon, Palisades Fire, benthic macroinvertebrates, tidewater goby, bioassessment, critical habitat, GIS

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Introduction

Wildfire activity in southern California has grown larger, more frequent, and more severe as a result of climate change and historical land management (Williams et al., 2019). Although fire is often studied as a terrestrial disturbance, its effects extend downstream. Loss of vegetation reduces slope stability, and hydrophobic soil layers limit infiltration, increasing runoff, erosion, and the delivery of sediment, ash, and contaminants to streams and estuaries (Gomez Isaza et al., 2022; Moazeni & Cerdà, 2024). The direction and duration of these aquatic effects depend on watershed conditions, the timing and intensity of post-fire rainfall, and when sampling occurs (Minshall, 2003; Rehn et al., 2011).

The Palisades Fire ignited on January 7, 2025, and burned a total of 23,448 acres in the Santa Monica Mountains, including much of the Topanga Creek watershed (CAL FIRE, 2026). Topanga Creek drains into Topanga Lagoon, which the U.S. Fish and Wildlife Service designates as critical habitat for the federally endangered northern tidewater goby, *Eucyclogobius newberryi* (USFWS, 2013). Tidewater gobies occupy coastal lagoons and the lower reaches of connected freshwater streams, and they feed largely on small benthic invertebrates (Swift et al., 1989; USFWS, 2005). Changes to the lagoon's benthic community may therefore affect both the goby's habitat and its food supply.

Benthic macroinvertebrates (BMIs) are central to this relationship. They mediate nutrient cycling, support food web stability, and form a primary prey base for the tidewater goby (Swift et al., 1989). Because BMI taxa differ in their tolerance to sedimentation and pollution, the

composition of a BMI assemblage is also a widely used indicator of aquatic ecosystem condition (Ode et al., 2016).

This study evaluated the hypothesis that post-fire changes to substrate and water quality in Topanga Lagoon would reduce BMI taxon richness and diversity. Our objectives were to (1) quantify changes in BMI richness and Shannon diversity before and after the Palisades Fire, (2) describe the response of selected indicator and pioneer taxa, and (3) compare BMI trends with tidewater goby survey estimates over the same period. We also describe a planned GIS study motivated by field observations of gobies beyond the mapped critical habitat boundary.

Methods

Study Area

Topanga Lagoon is a seasonally closed coastal lagoon at the mouth of Topanga Creek in Los Angeles County, California (Figure 1).

Figure 1

Topanga Canyon Lagoon Before and After Recovery From the Palisades Fire



Note. Left: the Topanga Canyon watershed and coastline immediately after the Palisades Fire (top photo: RCDSMM; bottom photo: Wray Sinclair). Right: Topanga Canyon Lagoon 20 months after the fire.

Field Sampling

Resource Conservation District of the Santa Monica Mountains (RCDSMM) staff collected substrate samples following the Surface Water Ambient Monitoring Program (SWAMP) bioassessment procedure (Ode et al., 2016; Figure 2). Samples were collected on May 28, 2024 (pre-fire), and on April 9, May 5, September 3, and November 12, 2025, and February 4, 2026 (post-fire).

Laboratory Processing

Samples were preserved in 70% ethanol and processed in a wet laboratory by CSU Channel Islands Summer Undergraduate Research Fellows (Figure 2). Specimens were sorted from substrate, identified to the lowest practical taxonomic level, and counted under

approximately 4.5× magnification. Because identifications combined family- and order-level groups, we report taxon richness rather than species richness.

Figure 2

Field Collection and Laboratory Processing of BMI Samples



Note. Left: the student research team on a SWAMP data collection excursion with RCDSMM in Topanga Lagoon. Right: Devan Page and Chris Blackwell sorting and identifying BMI samples in the wet laboratory.

Data Analysis

Taxon richness was calculated as the number of distinct taxa per sampling event. Diversity was calculated with the Shannon-Weaver index:

$$H' = -\Sigma\, p_i \ln p_i$$

where S is the number of taxa, the sum runs from $i = 1$ to S, and p_i is the proportion of individuals belonging to taxon i. Simpson's diversity index was also calculated for each sampling event as a complementary measure of dominance. Pre- and post-fire comparisons are descriptive; no formal statistical test was applied.

Tidewater Goby Surveys

RCDSMM biologists estimated tidewater goby abundance with seine surveys and recorded life stage and reproductive condition (Figure 3). Gobies were removed from the lagoon on January 17, 2025, and returned on July 17, 2025. Goby and BMI trends were compared descriptively.

Figure 3

Tidewater Goby Seine Surveys in Topanga Lagoon



Note. Left: Tatyana Ramirez and Dr. Brenton Spies count gobies in a seine net to estimate the population. Top right: Chris Blackwell and Dr. Spies pull a seine net through Topanga Lagoon. Bottom right: an adult northern tidewater goby in a viewing tank.

Results

Richness and Diversity

The BMI community collapsed after the Palisades Fire. Taxon richness fell from 20 taxa in the pre-fire sample to as few as 2 in early post-fire monitoring, a 90% reduction. Shannon

diversity declined from 1.74 to 0.34, indicating a community dominated by very few taxa (Table 1; Figure 4). Later samples showed partial increases in richness and diversity, but neither returned to pre-fire levels by February 2026 (Figure 5).

Table 1

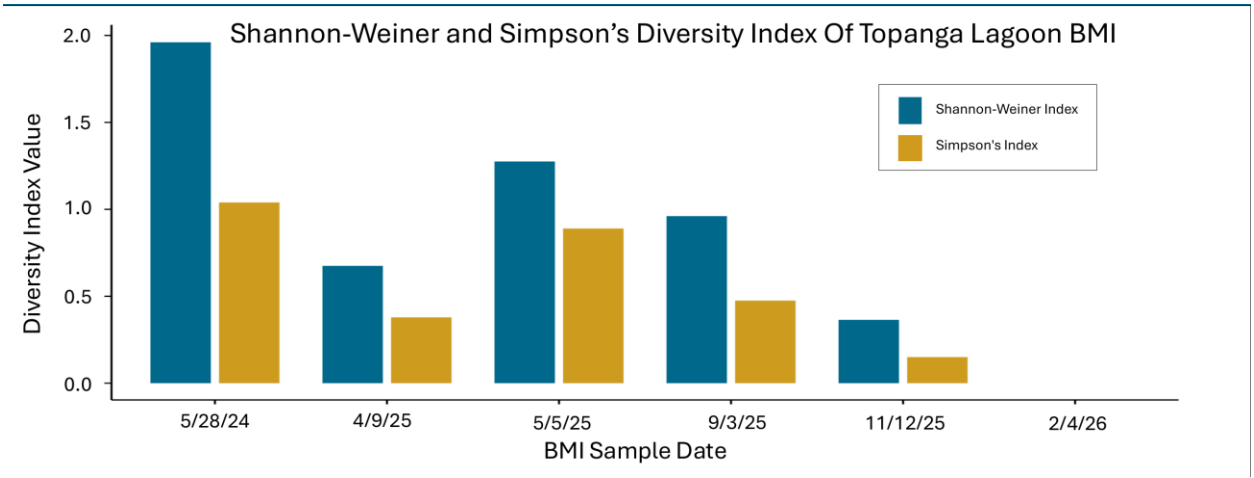
BMI Community Metrics Before and After the Palisades Fire

Measure	Pre-fire (May 28, 2024)	Post-fire
Taxon richness	20	Low of 2
Shannon diversity (H′)	1.74	Low of 0.34
Dominant early taxon	—	Ostracoda
Hydropsychidae	Present	Absent through Feb. 2026

Note. Post-fire values are the lowest observed across the five post-fire sampling events (April 2025–February 2026).

Figure 4

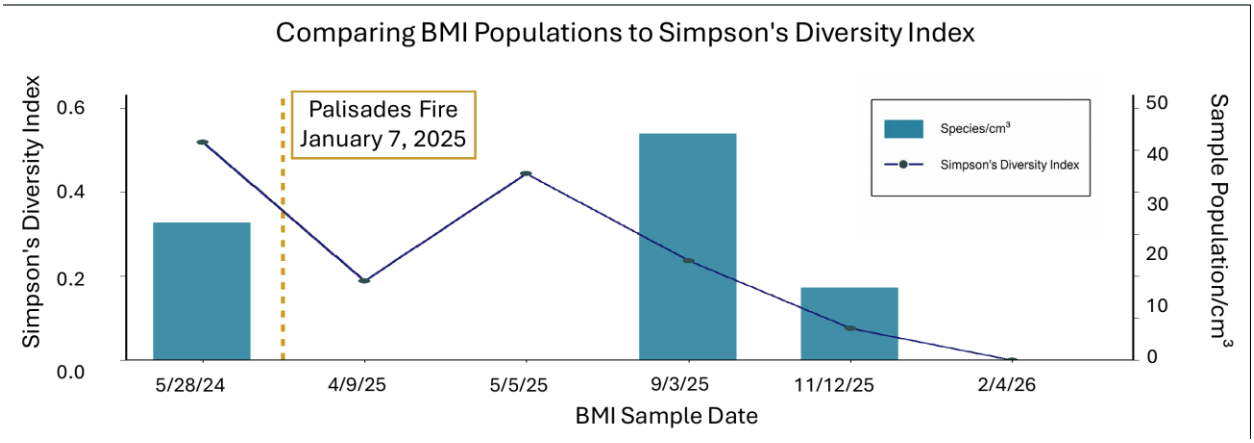
Shannon-Weaver and Simpson's Diversity Indices of Topanga Lagoon BMI Samples



Note. Shannon-Weaver values can range from 0 to about 4.3 and Simpson's values from 0 to 1; higher values indicate a more diverse community. The May 28, 2024 sample is pre-fire; all later samples are post-fire.

Figure 5

BMI Abundance Compared With Simpson's Diversity Index



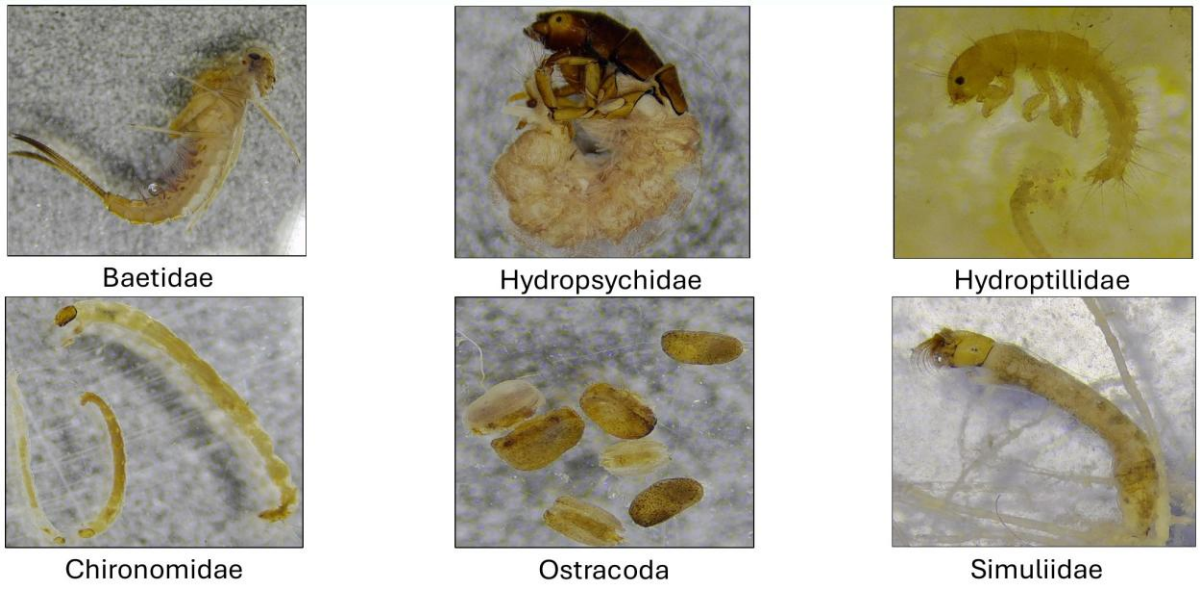
Note. Bars show the estimated number of individuals per cm³ in each sample; the line shows Simpson's diversity index. The dashed line marks the Palisades Fire on January 7, 2025.

Pioneer and Indicator Taxa

Ostracods dominated the first post-fire sample on April 9, 2025, roughly three months after ignition, making them the leading pioneer taxon of the recovering assemblage (Figure 6). Hydropsychidae (net-spinning caddisflies), which were present before the fire and are generally intolerant of fine sediment, were absent from every post-fire sample through February 2026.

Figure 6

Representative Benthic Macroinvertebrates From Topanga Lagoon Samples



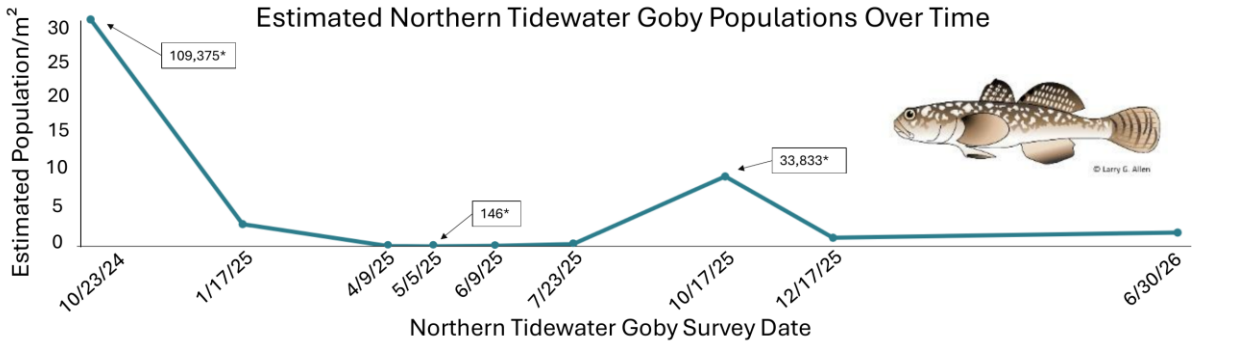
Note. Specimens photographed under magnification during laboratory processing: Baetidae, Hydropsychidae, Hydroptilidae, Chironomidae, Ostracoda, and Simuliidae.

BMI and Tidewater Goby Trends

Tidewater goby abundance followed a trajectory similar to the BMI community: a sharp post-fire decline, followed by a recovery spike beginning in fall 2025 (Figure 7). The goby record must be read alongside the January 17 removal and July 17 return of gobies to the lagoon, which shaped the population surveyed during this period.

Figure 7

Estimated Northern Tidewater Goby Population Over Time



Note. Values are estimated gobies per m²; labeled points give the actual estimated population size. Goby illustration © Larry G. Allen.

Discussion

The Palisades Fire was followed by a severe simplification of Topanga Lagoon's benthic community. A drop from 20 taxa to 2, with diversity falling by more than 80%, is consistent with the post-fire sediment pulses, substrate burial, and water quality changes reported in other burned California streams (Minshall, 2003; Rehn et al., 2011; Oliver et al., 2012). Several factors likely acted together, including post-fire runoff and sedimentation, seasonal salinity shifts, freshwater inflow, and lagoon breaching. Separating their effects will require more frequent sampling and paired water quality and habitat data.

The early dominance of ostracods fits the pattern of tolerant, fast-reproducing taxa colonizing disturbed habitat first, as generalist taxa did after wildfires in southern California streams (Rehn et al., 2011). Their abundance may have provided an important food resource during early recovery, although goby diet was not measured in this study. In contrast, the continued absence of Hydropsychidae more than a year after the fire suggests that sediment-sensitive taxa have not recovered, consistent with lasting declines in sensitive taxa after other

California fires (Oliver et al., 2012). Their return would be a useful marker of improving habitat condition in future monitoring.

The parallel decline and recovery of BMIs and gobies suggests a link between prey availability and goby population condition. However, the goby removal and return, together with differences in sampling dates between the two datasets, mean this relationship should be treated as a hypothesis rather than a demonstrated mechanism. Paired BMI and goby sampling, combined with goby diet analysis, would allow a direct test.

These results provide a baseline for tracking long-term recovery in a fire-affected coastal lagoon. As wildfire frequency increases across southern California, similar monitoring can help managers decide when habitat has recovered enough to support reintroduction or relocation of endangered species.

Proposed Follow-Up Capstone: Mapping Occupied Goby Habitat

During field surveys, our team observed tidewater gobies upstream of the boundary of critical habitat units. If confirmed, these observations suggest the current designation may not capture the full extent of habitat the species uses. A follow-up GIS study will test whether occupied reaches extend beyond current protection zones and identify the physical features that limit upstream access. The study will proceed in four steps:

1. **Compile occurrence data.** Assemble permitted goby survey records with precise coordinates, dates, methods, effort, and life stages, and compare them against the official USFWS critical habitat geometry and Federal Register boundary description after checking coordinate systems and positional accuracy.
2. **Model stream gradient.** Use LiDAR-derived elevation data to build a conditioned stream network and calculate reach-scale channel slope at several window lengths, since

the federal designation identifies steep gradient as one possible upstream limit (USFWS, 2013).

3. **Field-verify limiting features.** Survey candidate upper limits for passage barriers such as culverts and drops, dry reaches, flow permanence, substrate, and depth, with repeat surveys across seasons.
4. **Map and report.** Produce separate layers for observed occupancy, potentially suitable habitat, and designated critical habitat, with sensitivity analyses for elevation and slope estimates.

The resulting maps and technical report will be submitted to the U.S. Fish and Wildlife Service for consideration and prepared for peer-reviewed publication. Only USFWS, through its formal regulatory process, can revise a critical habitat designation; this study is intended to provide the spatial evidence for that review.

Limitations

This study has one pre-fire sampling event and no unburned reference site, so post-fire changes cannot be attributed to fire alone or separated from seasonal variation. Replication and sampling effort per event were limited, and BMI and goby surveys were not conducted on matching dates. Goby abundance was also affected by the removal and return of fish during the study period. Future monitoring should add replicate samples, a reference site, paired water quality and habitat measurements, and coordinated BMI and goby sampling.

Conclusion

The Palisades Fire was followed by a 90% loss of BMI taxa and a sharp decline in diversity in Topanga Lagoon, with only partial recovery more than a year later. Pioneer ostracods recolonized quickly, but sediment-sensitive caddisflies have not returned. Goby populations

followed a similar decline and recovery, suggesting that prey availability may matter for the species' post-fire recovery. Continued monitoring, together with the planned GIS analysis of occupied goby habitat beyond current habitat zones, can give managers the evidence needed to protect this endangered species in an increasingly fire-prone landscape.

Acknowledgments

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